

Visual Data Analysis Course Term Project Preparation

WS2025

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Choosing Data

Our data is based on the aircraft data from opensky-network.org. We plan to use this data to create a heatmap of flights either on a dedicated area or at a time interval.

Getting Data

We have stored our data in the [SeaFile](#). Our data is time-varying. The structure of our data is as follows:

```
[
  {
    "icao24": {icao24},
    "path":
    [
      {timestamp, lat, lon, altitude, heading, onground},
    ]
  },
]
```

Field	Type	Meaning
icao	string	24-bit ICAO aircraft address (hexadecimal), uniquely identifies the aircraft.
path	list<path_object>	Ordered list of trajectory samples. Each sample represents the aircraft state at one moment in time.

Field	Type	Meaning
timestamp	int	UNIX timestamp (seconds since epoch).
lat	float	Geographic latitude in decimal degrees.
lon	float	Geographic longitude in decimal degrees.
altitude	int	Altitude in meters (geometric altitude).
heading	int	Aircraft heading in degrees (0–359).
on_ground	boolean	true if aircraft is on the ground, false otherwise.

It is a spatial data with three dimensions + time.

Target Audience

- Flight enthusiasts
- Students

They can use this data to understand the route heatmaps across the globe. They may reason why do planes use some routes more than the others. They do not need prior experience in interpreting visualizations because it will be based on a earth map people see in their everyday life.